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Thermal brick support mounting bracket

Abstract

A thermal brick support system includes a mounting bracket and a shelf. The mounting bracket has a mounting plate and a pair of spaced apart legs extending from the mounting bracket. The legs extend perpendicular from the mounting plate and each have an extension portion extending from the mounting plate and a depending portion depending from the extension portion. The extension portion spaces the depending portion from a plane defined by the mounting plate to define a gap. The legs each have a hook formed therein on a front edge. The shelf has an upstanding leg and a transverse leg. The upstanding leg has through wall slots formed therein to cooperate with the mounting bracket hooks. When the shelf is positioned on the mounting bracket, the hooks engage the slots to support the shelf from the mounting bracket. A mounting bracket for a thermal brick support system is also disclosed.

Inventors:	Mason; Christopher (Hauppauge, NY)
Applicant:	Hohmann & Barnard, Inc. (Hauppauge, NY)
Family ID:	1000008780638
Assignee:	Hohmann & Barnard, Inc. (Hauppauge, NY)
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Primary Examiner: Cajilig; Christine T

Attorney, Agent or Firm: Andrus Intellectual Property Law, LLP

Background/Summary

BACKGROUND

- (1) The present disclosure relates to an improved mounting bracket, and more particularly, to a mounting bracket for a thermal brick support system.
- (2) Brick support systems for building are known. In one such system, a shelf is mounted to struts that are mounted to the building substrate. As such, the brackets rest on the substrate. The shelf is mounted to elongated brackets that are affixed to inserts in the substrate. The brackets have legs that extend outward to which the shelves are mounted. The system thus allows for mounting the shelf and the brick spaced from the substrate to accommodate insulation, vapor barriers and air gaps.
- (3) While such a system functions well for supporting and installing the brick façade, it can form a thermal bridge between the environs and the substrate. Such as thermal bridge allows for heat and cooling losses within the structure.
- (4) Accordingly, there is a need for a brick support system that reduces thermal losses from the substrate and structure. Desirably, such a system reduces thermal bridging. More desirably still, such a system allows for sufficient space between the brick and substrate to accommodate insulation, vapor barriers and air gaps.

SUMMARY

- (5) Various embodiments of the present disclosure provide a thermal brick support system. The system includes a mounting bracket having a mounting plate and a pair of spaced apart legs

extending from the mounting bracket. The legs extend perpendicular from the mounting plate, and each have an extension portion extending from the mounting plate and a depending portion depending from the extension portion. The extension portion spaces the depending portion from a plane defined by the mounting plate to define a gap. The legs each have a hook formed therein on a front edge thereof. In embodiments, the only connection between the legs is the mounting plate. The mounting bracket can be formed as unitary member.

(6) A shelf has an upstanding leg and a transverse leg. The upstanding leg has through wall slots formed therein to cooperate with the mounting bracket hooks. When the shelf is positioned on the mounting bracket, the hooks engage the slots to support the shelf from the mounting bracket.

(7) In embodiments, the hooks have an upwardly extending finger to secure the shelf on the mounting bracket. The hooks are formed on the front edge between an upper edge of the legs and a lower edge of the legs.

(8) The mounting plate can further include an opening therein. The opening can be a slotted opening.

(9) In embodiments, the mounting plate has a height and the legs have an overall height, and the height of the mounting plate is less than the overall height of the legs. The height of the mounting plate can be less than or equal to about $\frac{1}{2}$ of the overall height of the legs. The height of the mounting plate can be about 40% to 50% of the overall height of the legs.

(10) In embodiments, a depth of the mounting bracket is about $6\frac{1}{4}$ inches, a depth of the depending portion is about 5 inches and the gap is about $1\frac{1}{4}$ inches.

(11) In another aspect, a mounting bracket for a thermal brick support system for securing the system to a wall, includes a mounting plate that defines a plane and a pair of spaced apart legs extending from the mounting bracket. The legs extend perpendicular from the mounting plate and each have an extension portion extending from the mounting plate and a depending portion depending from the extension portion. The extension portion spaces the depending portion from a plane defined by the mounting plate to define a gap. The only connection between the legs is the mounting plate.

(12) The legs each have a hook formed therein on a front edge. When the mounting bracket is secured to the wall the depending portion is spaced from the plane defined by the mounting plate to define a gap between the depending portion and the wall.

(13) In embodiments, the mounting plate has a height and the legs have an overall height, and the height of the mounting plate is less than the overall height of the legs. The height of the mounting plate can be less than or equal to about $\frac{1}{2}$ of the overall height of the legs, and can be about 40% to 50% of the overall height of the legs.

(14) Other aspects, objectives and advantages will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an embodiment of a thermal brick support system;

(2) FIG. 2 is a perspective view of an embodiment of a mounting bracket for the thermal brick support system;

(3) FIGS. 3A and 3B are side and front views of the mounting bracket of FIG. 2;

(4) FIG. 4 is a side view similar to FIG. 3A showing a brick support shelf mounted to the mounting bracket;

(5) FIG. 5 is a sectional view showing the support system with the mounting bracket mounted to a wall or substrate and having brick courses supported by the system; and

(6) FIG. 6 is a side view of an embodiment of a brick overhead support member.

DETAILED DESCRIPTION

(7) While the present disclosure is susceptible of embodiments in various forms, there is described presently preferred embodiments with the understanding that the present disclosure is to be considered an exemplification and is not intended to limit the disclosure to the specific embodiments illustrated.

(8) Referring now to the figures and in particular to FIG. 1, there is shown an embodiment of a thermal brick support system **10**. The support system **10** includes a mounting bracket **12** and a shelf **14** supported on the mounting bracket **12**. The system **10** may also include overhead brick support members **16** having openings **18** therein and wires **20** for threading through the overhead members' openings **18** to support the bricks B. The mounting bracket **12** can be mounted to a structural form, such as a slab edge, steel beam, bond beam or the like. Other mounting arrangements will be understood by those skilled in the art. All such mounting arrangements are referred to collectively as wall, substrate, or wall or substrate and all such mounting arrangements are within the scope and spirit of the present disclosure.

(9) Referring now to FIGS. 2, 3A and 3B, the mounting bracket **12** includes a mounting plate **22**, and a pair of legs **24** extending from opposing ends **26** of the mounting plate **22**. An opening **28** in the mounting plate **22** is configured to receive a fastener F to fasten the mounting bracket **12** to the wall or substrate S. In embodiments, the opening **28** is a slotted opening. The mounting plate **22** defines a plane P.sub.22.

(10) In a current embodiment, the legs **24** extend about perpendicular from the mounting plate **22**. The legs **24** each include an extension portion **30** extending from the mounting plate **22** and a depending portion **32** extending downwardly or depending from the extension portion **30**. The extension portion **30** has a height h.sub.30 equal to the height h.sub.22 of the mounting plate **22**. That is, there is no rear wall of the depending portion **32**. The only rear wall is the mounting plate **22**, and the mounting plate **22** is the only connection between the legs **24**.

(11) The extension portion **30** is configured to space the depending portion **32** from the mounting plate plane P.sub.22 to define a gap **34** between the depending portion **32** and the wall or substrate S. In a current embodiment, the mounting plate **22** minimizes the surface area of the mounting bracket **12** that is in contact with the wall or substrate S and the extension portion **30** spaces the depending portion **32** from the wall or substrate S to minimize the thermal bridge between the mounting bracket **12** and the wall or substrate S and thus the building structure. In embodiments, the mounting bracket **12** is a unitary member.

(12) Upwardly extending hooks are **36** formed on a front edge **38** of the depending portions **32** of each of the legs **24** between an upper edge **40** of the legs **24** and a lower edge **42** of the legs **24**. A space **44** is formed between an upwardly oriented finger **46** at the end of each hook **36** and the depending portion **32** of its leg **24**.

(13) Referring now to FIGS. 1 and 4, the shelf **14** is formed as an L or angle having an upstanding leg **48** and a transverse leg **50**. The upstanding leg **48** includes through wall slots **52** formed between upper and lower edges **54**, **56** of the upstanding leg **48**. The slots **52** are configured to cooperate with the depending portion hooks **36**. When the shelf **14** is positioned on the mounting brackets **12**, the hooks **36** are inserted into the slots **52** and the fingers **46** rest on an inner wall **58** of the upstanding legs **48**, above the slots **52**. A lower portion **60** of the upstanding leg **48**, below the slots **52**, rests on the portion of the mounting bracket below the hooks **36**. This provides stability of the shelf **14** as positioned on the mounting bracket **12**.

(14) The shelf transverse leg **50** includes slots **62** formed therein. The slots **62** are configured to receive tabs **64** on the overhead brick support members **16**.

(15) In a current embodiment the overall height h.sub.12 of the mounting bracket **12** is about 14.125 (14 $\frac{1}{8}$) inches and the height h.sub.22 of the mounting plate **22** is about 6 inches or about 42.5 percent of the mounting bracket overall height h.sub.12. The gap or space **34** between the depending portion **32** and the rear of the mounting plate **22** or the wall or substrate S is about 1.25

(1¼) inches. The overall depth d.sub.12 of the mounting bracket **12**, exclusive of the hook **36** is about 6.25 (6¼) inches. The overall height h.sub.14 of the shelf **14** is about 6 inches and the depth d.sub.14 of the shelf **14** is about 4.75 (4¾) inches. The hook **36** extends from the leg depending portion **32** about 1.25 (1¼) inches and the space **44** between the interior wall of the hook **36** and the front edge **38** of the depending portion **32** (to receive the shelf **14**) is about 0.5 (½) inch.

(16) When the shelf **14** is installed on the mounting bracket **12**, the shelf **14** extends slightly below the depending leg lower edge **42**. In a current embodiment the shelf **14** extends about 0.25 (¼) inch below the lower edge **42**.

(17) Referring now to FIG. **5**, there is shown a sectional view of an example of a wall S having the thermal brick support system **10** mounted thereto and a number of courses of bricks B on the support system **10**. The mounting bracket **12** is mounted to the wall S by fasteners F, such as bolts. In the illustrated wall S, the mounting bracket **12** is mounted to a steel member. Insulation I is positioned around, above, below and between the mounting brackets **12** and in the space between the mounting bracket depending leg portions **32** and the wall S. A vapor barrier (not shown) can be installed over the insulation I and mounting brackets **12**, including the mounting bracket legs **24**, and the brick B installed over the vapor barrier. Other members, such as backing board D, condensation and waterproof barriers/films and the like may be installed as desired.

(18) It will be appreciated that the present thermal brick support system **10** provides advantages over known brick support systems. For example, the present system **10** allows for the installation of brick B on a wall or substrate S to include insulation I, a backing board D, a vapor barrier and an air gap between the brick B and the wall S. In addition, the present system **10** minimizes the thermal bridge created by contact of the steel members of the support system **10** (which is only the mounting plate **22**) with the wall or substrate S by reducing the surface area of the mounting bracket **12** that is in contact with the wall or substrate S. In addition, the present system **10** facilitates ready installation of brick B on the wall or substrate S using readily installed mounting brackets **12** with a hook **36** and slot **52** mounting configuration.

(19) It will be appreciated that the dimensions of the support system **10** and its components are examples only and that the system **10** and its components can have other dimensions as desired or required for a particular design or installation.

(20) In the present disclosure, the words “a” or “an” are to be taken to include both the singular and the plural. Conversely, any reference to plural items shall, where appropriate, include the singular. All percentages are percentages by weight, unless otherwise noted.

(21) All patents and published applications referred to herein are incorporated by reference in their entirety, whether or not specifically done so within the text of this disclosure.

(22) It will also be appreciated by those skilled in the art that the relative directional terms such as sides, upper, lower, top, bottom, rearward, forward and the like are for explanatory purposes only and are not intended to limit the scope of the disclosure.

(23) From the foregoing it will be observed that numerous modifications and variations can be effectuated without departing from the true spirit and scope of the novel concepts of the present disclosure. It is to be understood that no limitation with respect to the specific embodiments illustrated is intended or should be inferred. The disclosure is intended to cover by the appended claims all such modifications as fall within the scope of the claims.

Claims

1. A thermal brick support system, comprising: a mounting bracket having a mounting plate and a pair of spaced apart legs extending from the mounting bracket, the legs extending perpendicular from the mounting plate, the legs each having a front edge opposite the mounting plate, an extension portion extending from the mounting plate and a depending portion depending from the extension portion, the extension portion extending between an upper edge and an extension edge

with the depending portion extending away from the upper edge past the extension edge to a lower edge of the depending portion, the depending portion having a rear edge opposite the front edge and the extension edge spacing the depending portion from a plane defined by the mounting plate to define a gap between the plane and the rear edge, the legs each having a hook formed on the front edge at a position between the extension edge and the lower edge of the depending portion and extending away from the front edge; a shelf having an upstanding leg and a transverse leg, the upstanding leg having an upper edge, an inner wall surface on an included angle side of the shelf, and the upstanding leg having an outer wall surface opposite the inner wall surface and wall slots formed in the upstanding leg and configured to receive the mounting bracket hooks, wherein when the shelf is positioned on the mounting bracket, the hooks of the mounting bracket extend through the wall slots such that the outer wall surface of the shelf engages the front edge of the leg to support the shelf from the mounting bracket, with the upper edge at a position between the extension edge and the lower edge of the depending portion.

2. The system of claim 1, wherein the hooks have an upwardly extending finger defining an interior finger surface spaced apart from the front edge and the interior finger surface is configured to engage the interior wall surface of the upstanding leg to retain the shelf on the mounting bracket.

3. The system of claim 1, wherein the mounting plate further includes an opening therein.

4. The system of claim 3, wherein the opening is a slotted opening.

5. The system of claim 1, wherein the height of the mounting plate is less than or equal to about $\frac{1}{2}$ of the overall height of the legs.

6. The system of claim 5, wherein the height of the mounting plate is about 40% to 50% of the overall height of the legs.

7. The system of claim 1, wherein a depth of the mounting bracket is about 6- $\frac{1}{4}$ inches, a depth of the depending portion is about 5 inches and the gap is about 1 $\frac{1}{4}$ inches.

8. The system of claim 1, wherein the mounting bracket is a unitary member.

9. A mounting bracket for a thermal brick support system for securing the system to a wall, the mounting bracket comprising: a mounting plate defining a plane; and a pair of spaced apart legs extending from the mounting bracket, the legs extending perpendicular from the mounting plate, the legs each having a front edge opposite the mounting plate, an extension portion extending from the mounting plate and a depending portion depending from the extension portion, the extension portion extending between an upper edge and an extension edge with the depending portion extending away from the upper edge past the extension edge to a lower edge of the depending portion, the depending portion having a rear edge opposite the front edge and the extension edge spacing the depending portion from the plane defined by the mounting plate to define a gap between the plane and the rear edge, the legs each having a hook formed on the front edge at a position between the extension edge and the lower edge of the depending portion and extending away from the front edge.

10. The mounting bracket of claim 9, wherein the mounting plate has a height and wherein the legs have an overall height, and wherein the height of the mounting plate is less than the overall height of the legs.

11. The mounting bracket of claim 10, wherein the height of the mounting plate is less than or equal to about $\frac{1}{2}$ of the overall height of the legs.

12. The mounting bracket of claim 11, wherein the height of the mounting plate is about 40% to 50% of the overall height of the legs.

13. The system of claim 1, further comprising an overhead brick support member comprising a plate defining at least one opening therethrough, the at least one opening configured to receive at least one wire, the overhead brick support member depending from the transverse leg of the shelf.

14. The system of claim 13, wherein the overhead brick support member comprises a tab extending therefrom and the transverse leg of the shelf comprises a slot configured to receive the tab of the overhead brick support member.

15. The system of claim 14, wherein the tab of the overhead brick support member is a tee and the slot is oriented orthogonally to a final position of the overhead brick support member and configured to receive the tab therethrough wherein upon subsequent rotation, the tab an overhead brick support member is retained in the final position by the transverse leg of the shelf.

16. The system of claim 1, further comprising: an overhead brick support member comprising a plate defining at least one opening therethrough, and a tee shaped tab extending from the plate, wherein the tab is configured to be received within slot oriented parallel to the upstanding leg of the shelf, and the tab retains the overhead brick support member from the transverse leg in an orientation orthogonal to the transverse leg and the upstanding leg; and at least one wire received through the at least one opening in the overhead brick support member.

17. A thermal brick support system, comprising: a mounting bracket having a mounting plate and a pair of spaced apart legs extending from the mounting bracket, the legs extending perpendicular from the mounting plate, the legs each having an extension portion extending from the mounting plate and a depending portion depending from the extension portion, the extension portion spacing the depending portion from a plane defined by the mounting plate to define a gap, the legs each having a hook formed therein on a front edge thereof; and a shelf having an upstanding leg and a transverse leg, the upstanding leg having through wall slots formed therein to cooperate with the mounting bracket hooks, wherein when the shelf is positioned on the mounting bracket, the hooks engage the slots to support the shelf from the mounting bracket; wherein a depth of the mounting bracket is about $6\frac{1}{4}$ inches, a depth of the depending portion is about 5 inches and the gap is about $1\frac{1}{4}$ inches.
